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300218(15)

B. E. (First & Second Semester) Examination, 2020

(New Scheme)

(AEI, Bio Tech, Chem., Civil, CSE, Elect., EEE, EI, Et&T, IT, Mech., Mining, Metallurgy, Mechatronics, Production, Automobile Agriculture Branch)

APPLIED PHYSICS (New)

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each question is compulsory and carries 2 marks. Attempt any* ***two*** *parts from (b), (c) and (d) of each question which carry 7 marks each. Assume data wherever necessary.*

Unit-I

1. (a) According to relativity, the length of a rod in motion : 2

(i) is same as its rest length

(ii) is more than its rest length

(iii) is less than its rest length

(iv) None of these

(b) Describe Michelson-Morley experiment with the help of neat diagram and discuss its negative result. 7

(c) State the fundamental postulates of special theory of relativity and deduce Lorentz transformation equation of space and time. 7

(d) (i) At what speed will the mass of a body is 2·25 times its rest mass? 3

(ii) A particle of rest mass m_0 moves with speed $c/\sqrt{2}$. Calculate its mass, momentum, total energy and kinetic energy. 4

Unit-II

2. (a) Energy released per fission reaction is : 2

(i) $6{\cdot}2 \times 10^{-11}$ J

(ii) $4{\cdot}2 \times 10^{-11}$ J

(iii) $5{\cdot}2 \times 10^{-11}$ J

(iv) $3{\cdot}2 \times 10^{-11}$ J

(b) What is a mass spectrograph? Describe the construction and working of Bainbridge mass spectrograph. Show that its mass scale is linear. 7

(c) What is meant by nuclear fusion? Explain on the basis of Proton-Proton (P-P) cycle and carbon-nitrogen (C-N) cycle. 7

(d) (i) Electrons are accelerated through a potential difference of 200 V and then projected at right angles into a magnetic field of 0·01 wb/m^2. Calculate the velocity of the electrons on entering the field and the radius of the path. 3

(ii) Calculate the power output of a nuclear reactor which consumes 25 gm of U-235 per day. Assume 5% reactor efficiency. Energy released per fission is 200 MeV. 4

Unit-III

3. (a) Define Absorption coefficient of sound energy for a material. What is its units? 2

(b) Describe the Huygen's eyepieces and explain with the help of a diagram its working and position of its cardinal points. 7

(c) What are Ultrasonic Waves? Explain magneto striction method of producing ultrasonic waves. 7

(d) (i) Classroom has dimensions $20 \times 15 \times 15\ m^3$. The reverberation time is 3·5 sec. Calculate the total absorption of its surfaces and average absorption coefficient. 4

(ii) An ultrasonic source of 0·07 MHz sends down a pulse towards the seabed which returns after 0·65 sec. The velocity of sound in sea water is 1700 m/s. Calculate the depth of sea and the wavelength of pulse. 3

Unit-IV

4. (a) The diameter D_n of the nth dark Newton's ring is given as : 2

(i) $D_n^2 = (2n+1)\lambda R$

(ii) $D_n^2 = 4\lambda R_n$

(iii) $D_n^2 = 2(2n+1)\lambda R$

(iv) $D_n = 4\lambda R/n$

(b) What is Fresnel's Biprism? Explain how fresnel's biprism experiment can be used to determine the thickness of a thin sheet of transparent material. 7

(c) What is Reyleigh criterion and hence deduce an expression for resolving power of grating. 7

(d) (i) Newton's rings are observed normally in reflected light of wavelength $5{\cdot}9 \times 10^{-5}$ cm. The diameter of the 10th ring is 0·50 cm. Find the radius of curvature of the lens and the thickness of the film. 4

(ii) Find the minimum number of lines in a plane diffraction grating required to just resolve the sodium doublet (5890 Å and 5896 Å) in the (a) first order and (b) second order. 3

Unit-V

5. (a) The ratio of Einstein's coefficient A and B can be expressed as : 2

(i) $8\pi hc^3/v^3$

(ii) $8\pi hv/c$

(iii) $8\pi h\nu^3/c^3$

(iv) $8\pi hc/\nu$

(b) Explain in brief the characteristics of a laser beam. Describe with a neat sketch, working of Nd : YAG laser. 7

(c) What is the structure of an optical fibre? Explain the principle of propagation of light within a fibre and give expression for acceptance angle and Numerical aperture. 7

(d) (i) A laser beam can be focussed on an area equal to the square of its wavelength (λ^2). For a He-Ne laser, $\lambda = 6328$ Å. If the laser. 4

(ii) A step index fibre in air has a NA of 0·16, a core refractive index of 1·45 and a core diameter of 60 cm. Determine the normalised frequency for the fibre when light at a wavelength of 0·9 μ m is transmitted. 3